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Questions

Q1.

This question is about compounds containing chlorine.

(a) A precipitate of silver chloride is formed when silver nitrate solution reacts with sodium chloride solution.

A student wrote an ionic equation for the reaction.

$$Ag^{2+}(l) + 2Cl^{-}(l) \rightarrow AgCl_2(s)$$

Explain why this equation is incorrect, even though it is balanced.

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(b) A sample of a compound is analysed and found to contain **only** 3.09 g carbon, 0.26 g hydrogen and 9.15 g chlorine.

The molar mass of the compound is 97.0 g mol^{-1}.
Calculate the molecular formula of this compound.
You **must** show your working.

(3)

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(c) Nitrogen trichloride has the formula NCl_3.

(i) A sample of nitrogen trichloride contained only nitrogen atoms with mass number 14, and chlorine atoms with mass numbers 35 and 37.
Give the formula and mass/charge ratio for each of the **four** ions responsible for the molecular ion peaks in the mass spectrum of nitrogen trichloride.

(2)

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(ii) Complete the table to predict the shape and Cl—N—Cl bond angle in nitrogen trichloride.

(3)

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Number of bonding pairs of electrons on nitrogen	
Number of lone pairs of electrons on nitrogen	
Shape of molecule	
Cl—N—Cl bond angle	

(d) Aluminium chloride exists as an ionic lattice in the solid state and as a covalent dimer, Al_2Cl_6, in the gas phase, just above its boiling temperature.

(i) Explain why aluminium chloride in the solid state has significant covalent character.

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(ii) Describe how two $AlCl_3$ molecules are joined together in the dimer.
Include a diagram in your answer.

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(Total for question = 14 marks)

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Q2.

Which species is **not** tetrahedral?

- ☒ **A** CCl_4
- ☒ **B** CH_4
- ☒ **C** ICl_4^-
- ☒ **D** NH_4^+

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(Total for question = 1 mark)

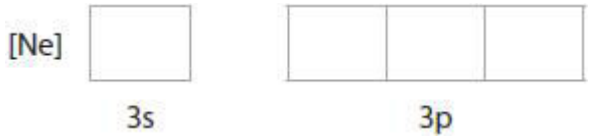
Q3.

This question is about phosphorus and some of its compounds.

(a) The atomic number of phosphorus is 15.

(i) Complete the electronic configuration of a phosphorus atom using the electrons-in-boxes notation.

(1)



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(ii) Explain why the first ionisation energy of phosphorus is greater than that of sulfur.

(2)

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(b) Phosphorus has a melting temperature of 44 °C.

Silicon has a melting temperature of 1410 °C.
Explain why the melting temperature of phosphorus is much lower than that of silicon.

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(c) Phosphorus(V) chloride, PCl_5 , exists as covalent molecules in the gaseous state.

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(i) Complete the table for a PCl_5 molecule.

(3)

Number of bonding pairs of electrons on phosphorus		
Number of lone pairs of electrons on phosphorus		
Shape of molecule		
Cl—P—Cl bond **angles**		

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(ii) In the solid state, phosphorus(V) chloride is ionic.
The cation and anion each have one phosphorus atom but a different number of chlorine atoms.
The cation is tetrahedral and the anion is octahedral.
Predict the formula of each ion. Include the charge on each ion.

(2)

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(d) In an experiment, 8.00 cm^3 of 0.250 mol dm^{-3} sodium hydroxide, NaOH, reacted completely with 10.0 cm^3 of 0.100 mol dm^{-3} phosphoric acid, H_3PO_4 .

Use these data to deduce the balanced equation for this reaction.
You **must** show your working.

(3)

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(e) Hydrated magnesium phosphate has the formula $Mg_3(PO_4)_2.\mathbf{y}H_2O$.

A sample of this compound contains 78.5% by mass of anhydrous magnesium phosphate.
Deduce the value of **y**.
You **must** show your working.
[Molar mass of anhydrous magnesium phosphate, $Mg_3(PO_4)_2$ = 262.9 g mol^{-1}]

(2)

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(Total for question = 16 marks)

Q4.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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How many lone pairs of electrons are there in the outer shell of the sulfur atom in a molecule of H_2S?

- ☐ **A** zero
- ☐ **B** one
- ☐ **C** two
- ☐ **D** four

(Total for question = 1 mark)

Q5.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

In which sequence are the molecules in order of **decreasing** bond angle?

- ☐ **A** $BeCl_2 > BCl_3 > CH_4$
- ☐ **B** $BeCl_2 > NH_3 > CH_4$
- ☐ **C** $CH_4 > BCl_3 > BeCl_2$
- ☐ **D** $CH_4 > NH_3 > BeCl_2$

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(Total for question = 1 mark)

Q6.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which molecule is planar?

- ☐ **A** CF_4
- ☐ **B** C_2F_4

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☒ **C** PF_5
☒ **D** SF_6

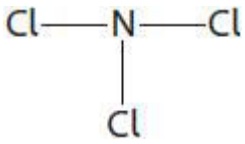
(Total for question = 1 mark)

Q7.

This question is about the structure and bonding of Group 5 chlorides.

(a) Nitrogen trichloride, NCl_3 , has a molecular structure.

The displayed formula of a molecule of NCl_3 is shown.



Complete the table for this molecule.

(3)

Number of bond pairs around N atom	
Number of lone pairs around N atom	
Cl-N-Cl bond angle	
Name of shape of molecule	

(b) Under standard conditions, phosphorus(V) chloride (PCl_5) is a solid made up of PCl_4^+ cations and PCl_6^- anions.

Antimony(V) chloride ($SbCl_5$) is a liquid made up of $SbCl_5$ molecules.

(i) Explain why PCl_5 has a higher melting temperature than $SbCl_5$.

(2)

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(ii) Draw a dot-and-cross diagram to show the bonding in a molecule of $SbCl_5$.
Use dots (•) to represent the Sb electrons, and crosses (**x**) to represent the Cl electrons. Show outer electrons only.

(2)

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(c) At low temperatures, $SbCl_5$ converts to Sb_2Cl_{10} which contains dative covalent bonds.

(i) State what is meant by the term dative covalent bond.

(1)

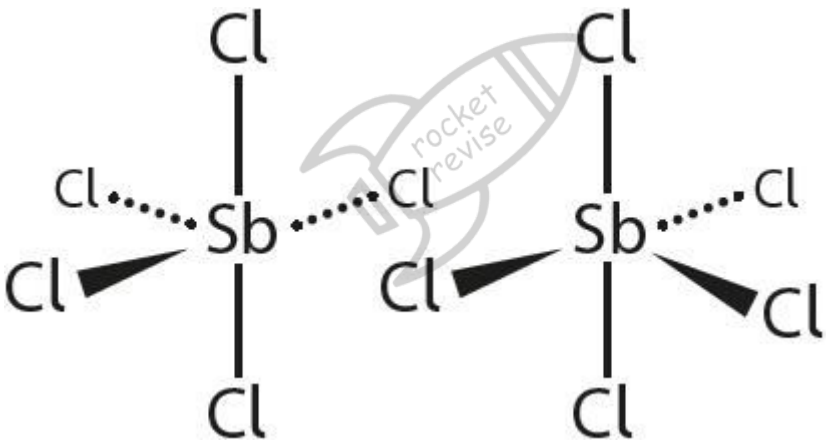
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(ii) Complete the diagram to show the dative covalent bonds in Sb_2Cl_{10} .

(1)



(d) Arsenic also forms a pentachloride with the formula $AsCl_5$.

Give **one** possible reason why nitrogen is the only Group 5 element that does **not** form a pentachloride.

(1)

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(Total for question = 10 marks)

Q8.

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This question is about the chlorides of beryllium and calcium.

(a) Complete the electronic configurations of the atoms of beryllium and calcium using the s, p, d notation.

(2)

Be $1s^2$

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Ca $1s^2$

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(b) In the gaseous state, beryllium chloride is molecular.

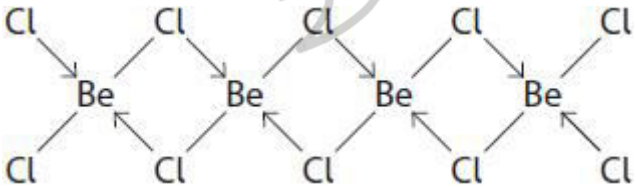
Draw a dot-and-cross diagram to show the bonding in a molecule of beryllium chloride, $BeCl_2$.

(2)

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(c) In the solid state, beryllium chloride forms a polymeric structure.

The diagram shows part of this structure.



The diagram uses lines and arrows to represent the two different types of covalent bond.
Describe how each type of bond is formed.

(2)

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(d) The Cl—Be—Cl bond angle is different in the two forms of beryllium chloride.

Predict the two bond angles, justifying your answers by referring to electron-pair repulsion theory.

(4)

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(e) Anhydrous calcium chloride is a crystalline, ionic solid which melts at 772 °C.

Draw a dot-and-cross diagram for calcium chloride.
Show the outer electrons only.

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(2)

(f) Explain why gaseous beryllium chloride and solid calcium chloride have different types of bonding.

(3)

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(Total for question = 15 marks)

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Q9.

This question is about the Group 4 element germanium and some of its compounds.

(a) Naturally-occurring germanium consists of five stable isotopes.

Explain what is meant by the term isotopes.

(2)

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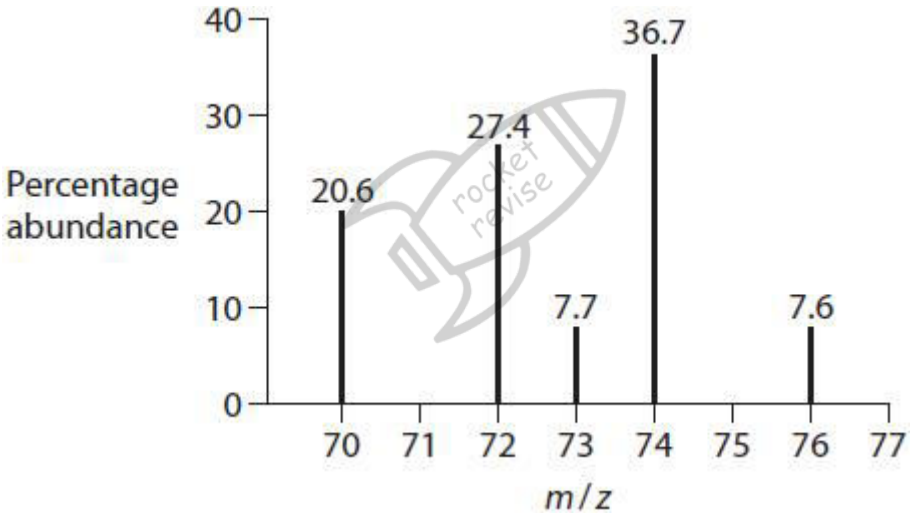
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(b) The mass spectrum of a sample of germanium is shown.



Calculate the relative atomic mass of this sample of germanium.

(2)

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(c) Give the electronic configuration of a germanium atom.

(1)

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(d) Germane is a compound with the formula GeH_4.

It can be formed by the reaction shown.

$$Na_2GeO_3 + NaBH_4 + H_2O \rightarrow GeH_4 + 2NaOH + NaBO_2$$

(i) Calculate the atom economy, by mass, for the formation of germane.
Use A_r of Ge = 72.6

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(2)

(ii) Germane can donate a proton to ammonia, forming the ion GeH_3^-.

$$NH_3 + GeH_4 \rightarrow NH_4^+ + GeH_3^-$$

Draw a dot-and-cross diagram of the ion GeH_3^-. Show the outer electrons only.

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(2)

(iii) Use your answer to (d)(ii) and electron-pair repulsion theory to predict the name of the shape and bond angle of the ion GeH_3^-

(2)

Name of the shape

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Bond angle

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(iv) Germane is toxic and the maximum permitted concentration is 0.640 mg m^{-3} in air.
Calculate the maximum number of germane molecules permitted in a laboratory with a volume of 231 m^3.
[Avogadro constant (L) = 6.02×10^{23} mol^{-1}]

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(4)

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(e) A halide of germanium, GeX_4, reacts with water as shown.

$$GeX_4 + 2H_2O \rightarrow GeO_2 + 4HX$$

1.50 g of GeX_4 was added to excess water.
The insoluble GeO_2 was removed by filtration.
The solution of hydrogen halide formed was then added to excess magnesium carbonate, forming 335.5 cm^3 of carbon dioxide at room temperature and pressure (r.t.p.).

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$$MgCO_3 + 2HX \rightarrow MgX_2 + CO_2 + H_2O$$

[Molar volume of a gas at r.t.p. = 24 000 $cm^3\ mol^{-1}$]

(i) Calculate the number of moles of carbon dioxide formed.

(1)

(ii) Deduce the number of moles of HX formed when GeX_4 reacted with the excess water.

(1)

(iii) Deduce the number of moles of GeX_4 that reacted with the excess water.

(1)

(iv) Calculate the molar mass of GeX_4 and hence identify X.

(2)

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(Total for question = 20 marks)

Q10.

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This question is about mass spectrometry and the shapes of molecules.

(a) In a mass spectrometer vaporised atoms are ionised, and the ions formed are accelerated, deflected and detected.

(i) State how atoms are ionised in the mass spectrometer.

(1)

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(ii) State how the ions formed are accelerated.

(1)

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(iii) Explain why isotopes of an element have the same chemical reactions but their ions are deflected differently in a mass spectrometer.

(2)

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(b) Data from mass spectra may be used to determine the relative atomic masses of elements.

(i) State what is meant by relative atomic mass.

(2)

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(ii) A sample of chlorine contains 75.53 % of ^{35}Cl and 24.47 % of ^{37}Cl.
Calculate the relative atomic mass of this sample of chlorine, giving your answer to **four** significant figures.

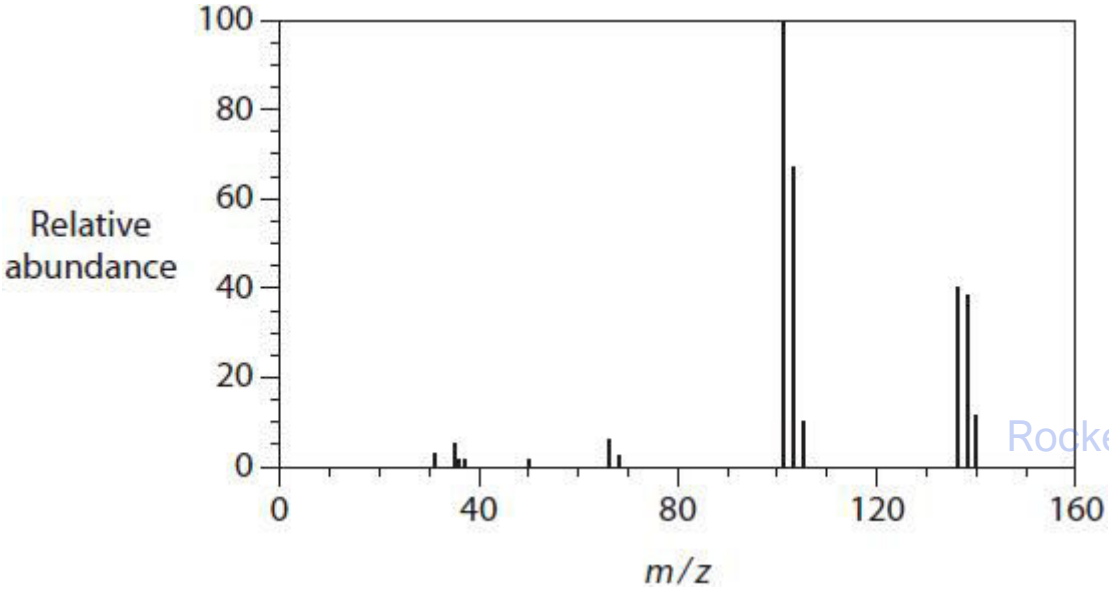
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(c) The mass spectrum of phosphorus trichloride, PCl_3, is shown.

Phosphorus has only one isotope, ^{31}P.



(i) There are three peaks in the region of m / z = 101–105.
Complete the table to show the ions responsible for these peaks.

(2)

m/z	Formula of ion	Relative peak height
101		9
103		6
105		1

(ii) Show that the relative peak heights given in the table are consistent with the isotopic ratio of ^{35}Cl to ^{37}Cl being 3 : 1.

(2)

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(d) (i) Draw a dot-and-cross diagram of a PCl_3 molecule.

Show outer electrons only.

(2)

(ii) Explain the shape of a PCl_3 molecule.

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(3)

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(Total for question = 17 marks)

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Q11.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which molecule has the largest bond angle?

- ☒ **A** BF_3
- ☒ **B** CF_4
- ☒ **C** H_2O
- ☒ **D** NH_3

(Total for question = 1 mark)

Q12.

This question is about the elements in Period 2 of the Periodic Table.

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(a) The first ionisation energies of some elements in Period 2 are shown.

Element	Li	Be	B	C	N	O	F	Ne
First ionisation energy / kJ mol^{-1}	520	900	801	1086	1402		1681	2081

(i) Explain the **general** trend in first ionisation energy across Period 2.

(2)

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(ii) Predict a value for the first ionisation energy of oxygen.
Justify your answer.

(3)

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(b) The second ionisation energy of lithium, Li, is 7298 kJ mol^{-1}.

(i) Write the equation for the second ionisation energy of lithium. Include state symbols.

(2)

(ii) Explain the large difference between the first and second ionisation energies of lithium.

(2)

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(c) Beryllium reacts with chlorine to form beryllium chloride.

In the gas phase, beryllium chloride exists both as a simple molecule, $BeCl_2$, and as a dimer, Be_2Cl_4.

(i) Draw a dot-and-cross diagram of the molecule, $BeCl_2$.

(1)

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(ii) Explain the shape and bond angle in the molecule $BeCl_2$.

(4)

Shape

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Bond angle

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Explanation

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(iii) Draw a diagram to show how two beryllium chloride molecules bond together to form the dimer, Be_2Cl_4, stating the type of bond involved.

(2)

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Type of bond

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(Total for question = 16 marks)

Q13.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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Which molecule has the **largest** bond angle?

- ☐ **A** BCl_3
- ☐ **B** CO_2
- ☐ **C** H_2O
- ☐ **D** NH_3

(Total for question = 1 mark)

Q14.

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What are the shapes of the $AlCl_3$ and PH_3 molecules?

	Shape of $AlCl_3$ molecule	Shape of PH_3 molecule
☐ **A**	pyramidal	pyramidal
☐ **B**	pyramidal	trigonal planar
☐ **C**	trigonal planar	trigonal planar
☐ **D**	trigonal planar	pyramidal

(Total for question = 1 mark)

Q15.

Boric acid is a white solid often used as an antiseptic.

(a) Boric acid contains 17.48% by mass of boron, 77.67% of oxygen and the remainder is hydrogen. The molar mass of boric acid is 61.8 g mol^{-1}.

[A_r values: H = 1 B = 10.8 O = 16]

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Show that the molecular formula of boric acid is H_3BO_3 .

You must show all your working.

(4)

(b) The formula of boric acid can also be written as $B(OH)_3$.

(i) Draw a dot-and-cross diagram for this molecule.

Show outer electrons only.

(3)

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(ii) Suggest a value for the O—B—O bond angle. Justify your answer.

(2)

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(Total for question = 9 marks)

Q16.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

This question is about bond angles.

(a) What is the expected bond angle in a boron trichloride molecule, BCl_3?

(1)

- ☐ **A** 90°
- ☐ **B** 107°
- ☐ **C** 109.5°
- ☐ **D** 120°

(b) What is the expected bond angle in an ammonium ion?

(1)

- ☐ **A** 90°
- ☐ **B** 104.5°
- ☐ **C** 107°
- ☐ **D** 109.5°

(c) What is the expected bond angle in an oxygen difluoride molecule, OF_2?

(1)

- ☐ **A** 90°
- ☐ **B** 104.5°
- ☐ **C** 109.5°
- ☐ **D** 180°

(Total for question = 3 marks)

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Q17.

Boron, aluminium and thallium are in Group 3 of the Periodic Table.

These elements form molecular compounds with chlorine with the formulae BCl_3, $AlCl_3$ and $TlCl_3$. The shape of these molecules depends on the electronic structures of the Group 3 elements.

(a) (i) Give the electronic configuration of aluminium.

(1)

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(ii) Compare and contrast the electronic structures of boron, aluminium and thallium.

(2)

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(iii) Deduce, using electron-pair repulsion theory, the expected shape of BCl_3, $AlCl_3$ and $TlCl_3$. Justify your answer.

(3)

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(b) Aluminium chloride is a solid at room temperature.

At the relatively low temperature of 453 K it sublimes.

(i) A sample of 5.00 g of aluminium chloride was heated to 455 K at a pressure of 1.01×10^5 Pa. When all the aluminium chloride had vaporised, the final volume of gas was 700 cm^3. Show that the data is consistent with the formula of aluminium chloride in the gas phase being Al_2Cl_6.
[Gas constant R = 8.31 J mol^{-1} K^{-1} Ideal gas equation $pV = nRT$]

(3)

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(ii) Draw the dot-and-cross diagram for Al_2Cl_6. Use dots (•) for the electrons of aluminium and crosses (×) for the electrons of chlorine.

(3)

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(c) Thallium also forms ions containing chlorine, for example the $TlCl_4^{3-}$ ion.

In this ion, the thallium atom has 10 electrons in its outermost shell.
Phosphorus in phosphorus pentachloride, PCl_5, also has 10 electrons in its outer shell.
Draw the shape of the $TlCl_4^{3-}$ ion and predict the bond angles.
Include any lone pairs of electrons that influence the shape.

(3)

(Total for question = 15 marks)

Q18.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Which molecule is polar?

- ☐ **A** $CO_2(g)$
- ☐ **B** $CCl_4(g)$
- ☐ **C** $BeCl_2(g)$

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☒ **D** $NH_3(g)$

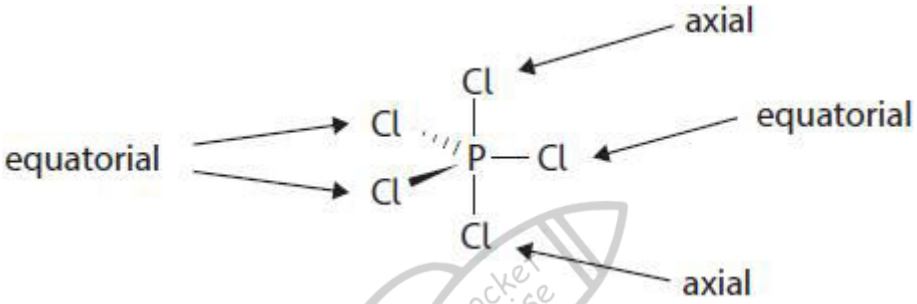
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Q19.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

In the gas phase, PCl_5 exists as molecules. They are trigonal bipyramidal.

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Each chlorine atom can be replaced one at a time by a fluorine atom.
The two axial chlorine atoms are replaced before the three equatorial atoms.

Which of the resulting molecules does **not** have a permanent dipole?

☒ **A** PCl_4F
☒ **B** PCl_3F_2
☒ **C** PCl_2F_3
☒ **D** $PClF_4$

(Total for question = 1 mark)

Q20.

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Sulfur hexafluoride, SF_6, is used in the electric power industry as an insulating gas.
It can be used in this way as it is very unreactive.

Fluorine reacts with sulfur to form SF_6.
Impurities of S_2F_{10} and SF_4 also form and must be removed because they are too reactive.

(a) Give the name of the compound S_2F_{10}.

(1)

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(b) Molecules of S_2F_{10} consist of two SF_5 units linked by a S—S bond.

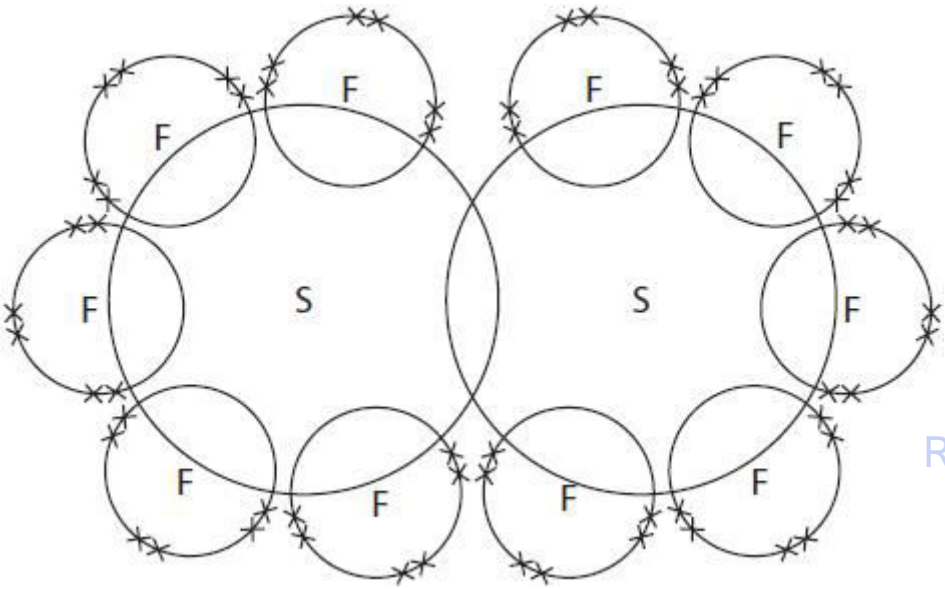
Complete the dot-and-cross diagram of S_2F_{10}, showing the outer shell electrons. Use dots (●) for the

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sulfur electrons and crosses (×) for the fluorine electrons.

(2)



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(c) Draw a dot-and-cross diagram of SF_4, showing all the outer shell electrons. Use dots (●) for the sulfur electrons and crosses (×) for the fluorine electrons.

(2)

(d) Complete the table showing clearly the shapes of the other two molecules and the F—S—F bond angles.

(4)

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Molecule	Diagram	F—S—F bond angle
SF_6		90°
S_2F_{10}		
SF_4		

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(e) Justify the shape around the sulfur atom in both SF_6 and SF_4.

(3)

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(f) SF_4 and S_2F_{10} are reactive, but SF_6 is unreactive.

Suggest reasons for this by considering the shape and bonding in each molecule.

(3)

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(Total for question = 15 marks)

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